

PARCS-Agent Benchmark

Task Dataset for Evaluating Distributed AI Agents

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2026

Abstract. This document defines 15 benchmark tasks for evaluating PARCS-enabled AI agents against standard sequential agents. Every task is fully self-contained: agents receive only the task prompt and access to the PARCS MCP server. Six tasks use closed-form mathematical parameters (no data transfer). Six tasks provide pre-generated input data via a single HuggingFace dataset URL passed as `datasetUrl` to `run_layer`. Three tasks require a two-layer pipeline where one layer loads a file and passes it to the next layer via `previousLayerResultJson`. All datasets and reference answers are published at **parcs-benchmark/parcs-agent-benchmark** on HuggingFace. Tasks span quantitative finance, machine learning, bioinformatics, combinatorial optimisation, epidemiology, cryptography, atmospheric science, and actuarial science.

Data Access Strategies

Strategy	Description	Tasks
Mathematical parameters	All data is defined by closed-form formulas in the prompt. Workers generate everything internally from the given seeds — no files, no downloads.	1, 2, 4, 9, 12, 14
HuggingFace dataset	Input data is loaded by passing a single HuggingFace raw URL as <code>datasetUrl</code> to <code>run_layer</code> . The MCP server downloads it once; all workers read from the shared NFS path in <code>input.DatasetPath</code> .	3, 5, 6, 7, 11, 15
Two-layer pipeline	Requires two files. Layer 1 (1 worker) loads the smaller file via <code>datasetUrl</code> and passes it via <code>resultJson</code> . Layer 2 (parallel workers) loads the larger file as <code>datasetUrl</code> and reads Layer 1 output from <code>previousLayerResultJson</code> .	8, 10, 13

Evaluation Setup

Agent A (PARCS-enabled): Has access to PARCS MCP tools (*get_cluster_info*, *create_session*, *run_layer*) and can fan work out across up to 21 parallel C# workers on a GKE cluster via Pub/Sub + KEDA autoscaling.

Agent B (baseline): No cluster access. Must compute everything sequentially. Given the same time budget, Agent B must either dramatically undersample, time out, or explicitly acknowledge incomplete results.

Both agents receive identical prompts. Primary metrics: **solution quality** (correctness and completeness), **time-to-result**, and **effective sample size** (fraction of required computation actually performed).

Dataset distribution: For tasks with pre-generated input data, Agent A passes the HuggingFace dataset URL as the *datasetUrl* parameter of *run_layer*. The MCP server downloads the file once and writes it to the shared cluster NFS volume; all daemon workers read it directly from *input.DatasetPath* — no per-worker download, no data transfer overhead.

Task Overview

#	Task	Domain	Data	Seq.	Par.
1	Monte Carlo Value-at-Risk for a Multi-Asset Portfolio	Quantitative Finance	Params	~20 min	~1 min
2	Barrier Option Price Surface via Monte Carlo	Derivatives Pricing	Params	~40 min	~2 min
3	Traveling Salesman — Parallel Simulated Annealing	Combinatorial Optimisation / Logistics	Dataset	~10 min (poor quality)	20 diverse runs
4	Epidemic SIR Parameter Sweep	Public Health / Epidemiology	Params	~30 min	~2 min
5	Hyperparameter Grid Search for Gradient Boosting	Machine Learning	Dataset	~60 min	~3 min
6	Bootstrap Confidence Intervals for Survival Analysis	Biostatistics / Clinical Research	Dataset	~20 min	~1 min
7	Protein Sequence All-vs-All Similarity	Bioinformatics / Drug Discovery	Dataset	~35 min	~2 min
8	Random Forest Construction from Scratch	Machine Learning	Pipeline	~20 min	~1 min
9	Safe Prime Generation and Primality Testing	Cryptography / Security Engineering	Pipeline	~15 min	~1 min
10	Financial Stress Testing — Historical Scenarios	Banking / Risk Management	Pipeline	~25 min	~2 min
11	Drug-Like Molecule Virtual Screening	Drug Discovery / Cheminformatics	Dataset	~15 min	~1 min
12	Insurance Collective Risk Model — Ruin Probability	Actuarial Science / Insurance	Params	~25 min	~2 min
13	Large-Scale Graph Betweenness Centrality	Network Science / Infrastructure Planning	Pipeline	~30 min	~2 min
14	Monte Carlo Radiative Transfer — Atmospheric Simulation	Atmospheric Science / Remote Sensing	Params	~30 min	~2 min
15	Parallel Genome k-mer Frequency Spectrum	Bioinformatics / Genomics	Dataset	~20 min	~1 min

Table 1. ‘Data’ column: how workers obtain the dataset. ‘Seq.’ = estimated wall-clock time for Agent B on one CPU core. ‘Par.’ = estimated wall-clock time for Agent A using 20 PARCS workers.

Task Definitions

Task 1	Monte Carlo Value-at-Risk for a Multi-Asset Portfolio	Quantitative Finance
• Agent Prompt	Estimate the 1-day 99% Value-at-Risk (VaR) and Conditional VaR (CVaR) for an equal-weight portfolio of 50 assets via Monte Carlo simulation. Setup: generate $A \in \mathbb{R}^{50 \times 50}$ with $N(0,1)$ entries from seed 42 (Box-Muller); compute $\Sigma = A A^T / 50 + 0.01 \cdot I$ and its lower Cholesky factor L . Simulate 2,000,000 scenarios total. Each worker draws its batch using $\text{seed} = \text{WorkerIndex} \cdot 1000 + 42$; for each scenario $z \sim N(0,1)^{50}$, portfolio loss $= -(1/50) \cdot \Sigma(L \cdot z)$. Aggregate across workers: $\text{VaR}_{99} = 99\text{th-percentile loss}$; $\text{CVaR}_{99} = \text{mean of losses above VaR}_{99}$. Return: {var_99, cvar_99}	
Formal Definition	Given a portfolio of 50 assets with weights $\mathbf{w} \in \mathbb{R}^{50}$ and a return covariance matrix $\Sigma \in \mathbb{R}^{50 \times 50}$, estimate the 1-day 99% Value-at-Risk and Conditional VaR by simulating 2,000,000 correlated market scenarios via Cholesky decomposition of Σ . VaR is the 1st percentile of the empirical loss distribution.	
• Mathematical parameters	No external dataset. Portfolio weights $\mathbf{w}_i = 1/50$ (equal-weight). Covariance matrix Σ generated once by each worker from seed = 42 as: $A = \text{randn}(50,50, \text{seed}=42)$; $\Sigma = A A^T / 50 + 0.01 \cdot I$. Each worker generates its 100,000 scenarios using new $\text{Random}(\text{WorkerIndex} * 1000 + 42)$ and the shared Cholesky factor L .	
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: benchmark (task_id=1)	
Parallelisation	Each worker simulates 100,000 of the 2,000,000 scenarios and returns its loss vector. A final single-worker layer pools all 20 loss vectors and computes percentiles.	
Applicability	Required daily by every major bank under Basel III internal models. Sequential simulation takes ~20 minutes; 20 parallel workers finish in ~1 minute.	
Success Metric	99% VaR, 99% CVaR, empirical loss distribution histogram.	
Timing	Sequential (Agent B): ~20 min Parallel 20× (Agent A): ~1 min	

Task 2	Barrier Option Price Surface via Monte Carlo	Derivatives Pricing
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• Agent Prompt	Price 200 down-and-out European call options on the grid $S \in \{80, 85, \dots, 125\}$, $\sigma \in \{0.10, 0.15, 0.20, 0.25\}$, $T \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ years. Parameters: $K=100$, $r=0.05$, barrier $B=0.85 \cdot S$. Use 500,000 GBM paths per grid point; each worker prices an equal slice of the 200 points using seed=WorkerIndex·999+7. Compute delta ($\Delta S=1$) and vega ($\Delta \sigma=0.01$) by central finite difference. Return: $\{[S, \sigma, T, \text{price}, \text{delta}, \text{vega}]\}$ for all 200 grid points.
Formal Definition	Price 200 down-and-out European call options across a grid of (S, σ, T) using geometric Brownian motion with 500,000 simulated paths per grid point. Barrier $B = 0.85 \cdot S$. Compute the full price surface and first-order Greeks (delta, vega) by central finite difference.
• Mathematical parameters	No external dataset. Grid defined by parameters: $S \in \{80, 85, 90, 95, 100, 105, 110, 115, 120, 125\}$, $\sigma \in \{0.10, 0.15, 0.20, 0.25\}$, $T \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ — 200 combinations in total. Risk-free rate $r = 0.05$. Strike $K = 100$. Each worker receives its 10 grid points via parametersJson and draws paths using new Random(WorkerIndex * 999 + 7).
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: benchmark (task_id=2)
Parallelisation	Each worker prices 10 grid points (200 / 20), running 500,000 paths per point. A final layer assembles the 200-entry price surface.
Applicability	Used by options market-makers to maintain real-time hedging tables. Sequential evaluation takes ~40 minutes; parallel ~2 minutes.
Success Metric	Price surface within 0.5% of Black-Scholes closed-form, full delta/vega grid.
Timing	Sequential (Agent B): ~40 min Parallel 20× (Agent A): ~2 min

Task 3	Traveling Salesman — Parallel Simulated Annealing	Combinatorial Optimisation / Logistics
• Agent Prompt	Find the shortest Hamiltonian tour through 300 cities using Simulated Annealing ($\alpha=0.9995$, $T_{\text{init}}=1000$, 500,000 iterations, 2-opt neighbourhood). Run multiple independent trials in parallel — each worker runs one trial with seed=WorkerIndex·777 and returns its best tour. City coordinates (city_id, x, y): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_03_cities.jsonl Pass this URL as datasetUrl to run_layer. Workers read via File.ReadAllText(input.DatasetPath!) and parse JSONL. Return: {best_tour_length, best_worker_index, all_trial_lengths:[float]}	
Formal Definition	Find the shortest Hamiltonian tour through 300 cities using Simulated Annealing with a 2-opt neighbourhood, geometric cooling $T(k) = T_0 \cdot \alpha^k$ ($\alpha = 0.9995$, 500,000 iterations). Run 20 independent trials with different random seeds; return the best tour found.	

• HuggingFace dataset	300 city coordinates pre-generated and published on HuggingFace. Fields: city_id (int), x (float), y (float). Pass the dataset URL as datasetUrl to run_layer; workers read it with File.ReadAllText(input.DatasetPath!) and parse JSONL. Each worker runs one independent SA trial with seed = WorkerIndex × 777.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_03_cities · benchmark (task_id=3)
Parallelisation	Each worker runs one independent SA trial. A final layer compares all 20 best-tour lengths and returns the global minimum.
Applicability	Underpins last-mile delivery routing, field service scheduling, PCB drilling. One sequential SA run takes ~3 minutes and explores a single trajectory; 20 parallel trials with different seeds find substantially better solutions.
Success Metric	Best tour length, convergence curve per seed, % improvement over nearest-neighbour.
Timing	Sequential (Agent B): 1 run (poor quality) Parallel 20× (Agent A): 20 diverse runs

Task 4 Epidemic SIR Parameter Sweep Public Health / Epidemiology	
• Agent Prompt	Simulate a discrete-time SIR epidemic on N=1,000,000 individuals for 365 days across a 20×20 parameter grid: $R_0 \in \text{linspace}(0.8, 4.0, 20)$, $\theta \in \text{linspace}(0.01, 0.20, 20)$. Model: $\beta = R_0/14$, $\gamma = 1/14$; when $I/N \geq \theta$, halve β . Initial state: S=999,900, I=100, R=0. Each worker simulates one row of the 20×20 grid (20 combinations). Return: [{R0, theta, peak_infected_frac, attack_rate, days_to_peak, intervention_days}] for all 400 combinations.
Formal Definition	Simulate a discrete-time SIR epidemic on N = 1,000,000 individuals over 365 days. Sweep a 20×20 grid: $R_0 \in [0.8, 4.0]$ and intervention threshold $\theta \in [0.01, 0.20]$ (fraction infected triggering a 50% contact-rate reduction). Compute peak infected fraction, total attack rate, days to peak, and intervention days for all 400 combinations.
• Mathematical parameters	No external dataset. Grid is fully specified by the sweep bounds above. Initial conditions: $S_0 = 999,900$, $I_0 = 100$, $R_0 = 0$. Recovery rate $\gamma = 1/14$. Each worker receives its 20 grid points via parametersJson (row index encoded as a string). The simulation is deterministic — no random seed required.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_04_sir_reference · benchmark (task_id=4)
Parallelisation	Each worker simulates 20 parameter combinations (one row of the 20×20 grid). A final layer assembles results into a heatmap.

Applicability	Public health agencies use sweeps to evaluate lockdown trigger policies. 400 simulations take ~30 minutes sequentially; 20 workers finish in ~2 minutes.
Success Metric	Full 20×20 heatmaps of peak infected %, attack rate, days to peak, intervention days.
Timing	Sequential (Agent B): ~30 min Parallel 20× (Agent A): ~2 min

Task 5 Hyperparameter Grid Search for Gradient Boosting Machine Learning	
• Agent Prompt	Evaluate all 20 gradient-boosting configurations on a binary classification dataset (80,000 rows, 25 features f0...f24, label) using 5-fold cross-validated AUC-ROC. Each worker evaluates one configuration (config_index=WorkerIndex). Training data: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_05_data.jsonl Pass this URL as datasetUrl. Workers read via File.ReadAllText(input.DatasetPath!). The 20 configurations are the ordered cross-product of: learning_rate ∈ {0.01, 0.05, 0.1} max_depth ∈ {3, 5, 7} n_estimators ∈ {100, 300} subsample ∈ {0.8, 1.0} Return: {best_config_index, best_auc_roc, all_configs:[{config_index, auc_roc}]}
Formal Definition	Train a gradient boosting classifier on a synthetic tabular dataset (80,000 rows, 25 features) for each of 20 hyperparameter configurations: learning rate ∈ {0.01, 0.05, 0.1}, max_depth ∈ {3, 5, 7}, n_estimators ∈ {100, 300}, subsample ∈ {0.8, 1.0}. Evaluate each with 5-fold cross-validation. Return the best configuration and its CV AUC-ROC.
• HuggingFace dataset	Training dataset (80,000 rows) pre-generated and published on HuggingFace. Fields: f0...f24 (float), label (0/1). Pass the dataset URL as datasetUrl to run_layer. The 20 configurations are the ordered cross-product of learning_rate∈{0.01,0.05,0.1}, max_depth∈{3,5,7}, n_estimators∈{100,300}, subsample∈{0.8,1.0}, indexed 0–19. Each worker evaluates one configuration (config_index = WorkerIndex).
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_05_data · task_05_configs · benchmark (task_id=5)
Parallelisation	Each worker trains one hyperparameter configuration (all 5 CV folds). A final layer ranks the 20 results by CV AUC-ROC.
Applicability	Hyperparameter optimisation dominates ML pipeline runtime in production. Sequential 5-fold CV across 20 configs takes ~60 minutes; parallel ~3 minutes.
Success Metric	Best CV AUC-ROC, ranked configuration table, training time per configuration.

Timing	Sequential (Agent B): ~60 min Parallel 20× (Agent A): ~3 min
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Task 6	Bootstrap Confidence Intervals for Survival Analysis	Biostatistics / Clinical Research
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• Agent Prompt	Fit a Cox proportional hazards model on 8,000 patients (covariates cov_0...cov_4, observed time, event indicator) and compute 95% bootstrap confidence intervals for all 5 hazard ratios using B=10,000 bootstrap resamples. Workers each run an equal share of resamples; use seed=WorkerIndex·1000+resample_index per resample. Patient data (cov_0...cov_4, time, event): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_06_patients.jsonl Pass this URL as datasetUrl. Workers read via File.ReadAllText(input.DatasetPath!). Return: {cov_0...cov_4: {ci_lower, ci_upper, point_estimate}}
Formal Definition	Fit a Cox proportional hazards model on a synthetic patient survival dataset (N = 8,000, time-to-event, binary event indicator, 5 covariates) and compute 95% bootstrap CIs for all hazard ratios using B = 10,000 resamples. Each resample draws N patients with replacement and fits the Cox model.
• HuggingFace dataset	Patient dataset (8,000 rows) pre-generated and published on HuggingFace. Fields: cov_0...cov_4 (float), time (float), event (0/1). True hazard ratios: $\beta = (0.5, -0.3, 0.8, 0.1, -0.6)$. Pass the dataset URL as datasetUrl to run_layer. Workers each run 500 bootstrap resamples with seed = WorkerIndex × 1000 + resample_index.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_06_patients · benchmark (task_id=6)
Parallelisation	Each worker runs 500 bootstrap resamples (10,000 / 20). Results — coefficient matrices — are concatenated; a final layer extracts percentiles.
Applicability	Accurate CIs are required for regulatory submission in Phase III clinical trials. Sequential B = 10,000 bootstrap takes ~20 minutes; parallel ~1 minute.
Success Metric	95% CI per hazard ratio, bootstrap distribution histograms, comparison to Wald CIs.
Timing	Sequential (Agent B): ~20 min Parallel 20× (Agent A): ~1 min

Task 7	Protein Sequence All-vs-All Similarity	Bioinformatics / Drug Discovery
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• Agent Prompt	Compute the full 600×600 pairwise similarity matrix for 600 protein sequences using simplified Smith-Waterman (match=+2, mismatch=-1, gap=-2). Normalise scores to [0,1] by dividing by 2·min(len_i,len_j). Workers each compute an equal block of the upper-triangular matrix (~9,000 pairs at parallelism=20). Identify the top-20 most similar pairs and count clusters with similarity>0.7. Protein sequences (seq_id, length, sequence): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_07_sequences.jsonl Pass this URL as datasetUrl. Workers read via File.ReadAllText(input.DatasetPath!). Return: {top_20_pairs:[{seq_i, seq_j, score}], n_clusters_above_0p7, matrix_checksum}
Formal Definition	Compute the full 600×600 pairwise similarity matrix for 600 synthetic protein sequences using a simplified Smith-Waterman local alignment scoring (match +2, mismatch -1, gap -2). Output the top-20 most similar pairs, cluster by average-linkage hierarchical clustering, and identify families with similarity > 0.7.
• HuggingFace dataset	600 protein sequences pre-generated and published on HuggingFace. Fields: seq_id (int), length (int), sequence (string, 20-letter AA alphabet). Pass the dataset URL as datasetUrl to run_layer. Workers each compute an equal block of the 179,700-pair upper-triangular similarity matrix (~9,000 pairs per worker at parallelism=20).
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_07_sequences · benchmark (task_id=7)
Parallelisation	Workers compute blocks of the 179,700-pair upper-triangular matrix (~9,000 pairs per worker). A final layer merges and ranks results.
Applicability	Used in drug discovery to identify protein homologs as alternative targets. Sequential computation of 180,000 alignments takes ~35 minutes; parallel ~2 minutes.
Success Metric	Full similarity matrix, top-20 pairs with scores, cluster dendrogram data.
Timing	Sequential (Agent B): ~35 min Parallel 20× (Agent A): ~2 min

• Agent Prompt	Train a random forest of 200 decision trees on a regression dataset and evaluate on a held-out test set. Tree config: bootstrap sample, $\sqrt{30} \approx 5$ features/split, max_depth=15, min_leaf_size=5. Use a two-layer pipeline: Layer 1 — workers each grow a subset of trees from the training data (tree seed=WorkerIndex·200+treeIndex): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_08_train.jsonl (60,000 rows, f0...f29, target) Layer 2 — one worker aggregates all trees and evaluates the ensemble: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_08_test.jsonl (10,000 rows) Pass each file as datasetUrl to its respective run_layer call. Return: {oob_rmse, test_rmse, top_10_feature_importances:[{feature, importance}]}
Formal Definition	Train a random forest of 200 decision trees on a synthetic regression dataset (60,000 rows, 30 features). Each tree uses a bootstrap sample, $\sqrt{30} \approx 5$ features per split, max depth = 15, min leaf size = 5. Report OOB RMSE, feature importance (mean decrease in impurity), and test predictions on 10,000 held-out rows.
• Two-layer pipeline	Training (60,000 rows) and test (10,000 rows) datasets pre-generated on HuggingFace. Fields: f0...f29 (float), target (float). Two-layer pipeline: Layer 1 workers grow trees from task_08_train.jsonl (tree seed = WorkerIndex × 200 + treeIndex); Layer 2 (1 worker) evaluates the ensemble using task_08_test.jsonl passed as datasetUrl.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_08_train · task_08_test · benchmark (task_id=8)
Parallelisation	Each worker grows 10 trees independently (200 / 20). A final layer averages all 200 trees' predictions and aggregates feature importance vectors.
Applicability	Random forests are production models in credit scoring, fraud detection, and insurance pricing. Growing 200 trees sequentially takes ~20 minutes; parallel ~1 minute.
Success Metric	OOB RMSE, test RMSE, feature importance ranking, agreement with reference.
Timing	Sequential (Agent B): ~20 min Parallel 20× (Agent A): ~1 min

• Agent Prompt	Among 10,000 randomly generated 512-bit odd integers, count how many pass a Miller-Rabin primality test ($k=20$ witness rounds) and how many are safe primes ($(p-1)/2$ also prime). Workers each test an independent batch using $\text{seed}=\text{WorkerIndex} \cdot 9973$ to generate candidates. Compare the empirical prime density to the prime number theorem prediction $1/\ln(2^{12})$. Return: {prime_count, safe_prime_count, empirical_density, pnt_density, iteration_histogram:[int]}
Formal Definition	Generate and test 10,000 candidate 512-bit odd integers for primality using Miller-Rabin with $k = 20$ witness rounds (false-positive probability $< 4^{-20}$). For each confirmed prime p , check whether $(p-1)/2$ is also prime (safe prime). Report counts, empirical prime density vs. prime number theorem prediction, and distribution of iterations.
• Mathematical parameters	No external dataset. Each worker generates its 500 candidate integers using $\text{new Random}(\text{WorkerIndex} * 9973)$ to produce 512-bit odd numbers. The Miller-Rabin witnesses are drawn from $\text{new Random}(\text{candidate_index})$ for reproducibility. Fully self-contained — no data transfer.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: benchmark (task_id=9)
Parallelisation	Workers each test 500 candidates. Embarrassingly parallel — no inter-worker communication. A final layer aggregates counts and statistics.
Applicability	Safe prime generation is the bottleneck in RSA and Diffie-Hellman key generation. Demonstrates throughput difference between sequential and cluster-based cryptography.
Success Metric	Prime count, safe-prime count, empirical density vs. PNT, iteration histogram.
Timing	Sequential (Agent B): ~15 min Parallel 20× (Agent A): ~1 min

**Task
10**

Financial Stress Testing — Historical Scenario Repricing

Banking / Risk
Management

• Agent Prompt	Reprice a portfolio of 300 financial instruments under 500 stress scenarios using delta-linear approximation: $P\&L;_s = \sum_i w_i \cdot (\delta_i \cdot shock_s)$. Identify the 10 worst-P&L scenarios and attribute losses to primary risk drivers. Use a two-layer pipeline: Layer 1 (1 worker) — load the portfolio and pass it via resultJson: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_10_portfolio.jsonl (instrument_id, weight, delta_0...delta_14) Layer 2 (parallel workers) — each reprices the portfolio under its share of scenarios; receive portfolio from previousLayerResultJson and load scenarios as datasetUrl: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_10_scenarios.jsonl (scenario_id, shock_0...shock_14) Return: {worst_10_scenarios:[{scenario_id, pnl}], pnl_var_99, pnl_std}
Formal Definition	Reprice a portfolio of 300 financial instruments under 500 stress scenarios (percentage shocks to 15 risk factors: equity indices, interest rates, FX, credit spreads) using delta-linear approximation. Identify the 10 worst P&L scenarios and attribute losses to primary risk-factor drivers.
• Two-layer pipeline	Portfolio (300 instruments) and scenario (500 shocks) data pre-generated on HuggingFace. Portfolio fields: instrument_id, weight, delta_0...delta_14. Scenario fields: scenario_id, shock_0...shock_14. Two-layer pipeline: Layer 1 (1 worker) loads the portfolio from task_10_portfolio.jsonl and passes it via resultJson; Layer 2 workers each price the portfolio under 25 scenarios from task_10_scenarios.jsonl (passed as datasetUrl).
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_10_portfolio · task_10_scenarios · benchmark (task_id=10)
Parallelisation	Each worker reprices the portfolio under 25 scenarios (500 / 20). A final layer assembles the P&L distribution and ranks scenarios.
Applicability	Required quarterly by regulators (FRTB, Basel III) for all trading-book portfolios at major financial institutions.
Success Metric	Full P&L distribution, worst-10 scenarios, driver attribution, stressed VaR at 99%.
Timing	Sequential (Agent B): ~25 min Parallel 20× (Agent A): ~2 min

• Agent Prompt	Score 50,000 drug-like molecules against a reference kinase inhibitor fingerprint using Tanimoto similarity on 2048-bit Morgan fingerprints. Proxy score= $0.7 \cdot \text{Tanimoto} + 0.1 \cdot (1 - \text{MW}/700) + 0.1 \cdot (1 - \text{HBD}/8) + 0.1 \cdot (1 - \log P/7)$. Apply Lipinski Rule-of-Five filters ($\text{MW} \leq 500$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$, $\log P \leq 5$). Workers each score an equal partition of the library. Reference fingerprint: 2048-bit vector from seed=0 (bit-set prob 0.05). Molecule library (mol_id, fingerprint as 512-char hex, mw, hbd, hba, logp): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_11_library.jsonl Pass this URL as datasetUrl. Workers read via <code>File.ReadAllText(input.DatasetPath!)</code> . Return: {top_100_mol_ids:[int], lipinski_pass_rate, mean_tanimoto}
Formal Definition	Score 50,000 molecules against a reference kinase inhibitor fingerprint using Tanimoto similarity on 2048-bit Morgan fingerprints. Apply Lipinski Rule-of-Five filters ($\text{MW} \leq 500$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$, $\log P \leq 5$). Compute a docking score proxy as a weighted sum of fingerprint similarity and 4 physicochemical descriptors. Return the top-100 ranked candidates.
• HuggingFace dataset	Library of 50,000 molecules pre-generated and published on HuggingFace. Fields: mol_id (int), fingerprint (512-char hex, 2048 bits), mw, hbd, hba, logp. Reference fingerprint: 2048-bit vector generated from seed=0 (bit-set prob 0.05). Pass the dataset URL as datasetUrl to run_layer. Workers each score an equal partition of the library (mol_ids WorkerIndex×2500 ... +2499 at parallelism=20).
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_11_library · benchmark (task_id=11)
Parallelisation	Workers each score 2,500 molecules (50,000 / 20). A final layer merges ranked lists, applies Lipinski filters, and returns the global top-100.
Applicability	Virtual screening reduces wet-lab funnel from millions of compounds to hundreds. Sequential scoring of 50,000 molecules takes ~15 minutes; parallel ~1 minute.
Success Metric	Top-100 candidate list, Tanimoto scores, filter-pass rate, score distribution.
Timing	Sequential (Agent B): ~15 min Parallel 20× (Agent A): ~1 min

**Task
12**

Insurance Collective Risk Model — Ruin Probability

Actuarial Science /
Insurance

• Agent Prompt	Simulate 2,000,000 annual insurance policy years under a compound Poisson risk process: claim count $N \sim \text{Poisson}(\lambda=200)$, severity $X \sim \text{LogNormal}(\mu=8, \sigma=1.5)$. Premium $P=1.2 \cdot E[S]$; initial surplus $U=500,000$. Workers each simulate an independent batch using $\text{seed_counts}=\text{WorkerIndex} \cdot 2053 + 1$, $\text{seed_severities}=\text{WorkerIndex} \cdot 3571 + 2$. Estimate: • Ruin probability within 1, 5, and 10 years • 99.5th-percentile annual aggregate loss (Solvency II SCR proxy) • Expected deficit given ruin Return: {ruin_prob_1yr, ruin_prob_5yr, ruin_prob_10yr, scr_995, expected_deficit_given_ruin}
Formal Definition	Simulate 2,000,000 policy years under a compound Poisson risk process: claim count $N \sim \text{Poisson}(\lambda = 200)$, severity $X \sim \text{Lognormal}(\mu = 8, \sigma = 1.5)$. Premium $P = (1 + \theta) \cdot E[S]$ with $\theta = 0.2$; initial surplus $U = 500,000$. Estimate probability of ruin within 1, 5, and 10 years; 99.5th-percentile annual loss (Solvency II SCR proxy); expected deficit given ruin.
• Mathematical parameters	No external dataset. All parameters above fully specify the process. Each worker simulates 100,000 policy years using new Random(WorkerIndex * 2053 + 1) for claim counts and new Random(WorkerIndex * 3571 + 2) for severities. Fully self-contained.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: benchmark (task_id=12)
Parallelisation	Each worker simulates 100,000 policy years (2,000,000 / 20). A final layer pools results and computes empirical ruin probabilities and percentiles.
Applicability	Used by insurers to determine required capital reserves under Solvency II. Sequential simulation at 2,000,000 runs takes ~25 minutes.
Success Metric	Ruin probabilities at 3 horizons, SCR estimate, time-to-ruin distribution, Cramér-Lundberg comparison.
Timing	Sequential (Agent B): ~25 min Parallel 20× (Agent A): ~2 min

• Agent Prompt	On a synthetic weighted road-network graph (5,000 nodes, ~25,000 edges, weights in minutes), compute Dijkstra shortest paths from 400 source nodes, approximate betweenness centrality, diameter and average path length (hops), and the top-20 most critical edges by removal impact. Use a two-layer pipeline: Layer 1 (1 worker) — load the node coordinate map and pass it via resultJson: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_13_nodes.jsonl (node_id, x, y) Layer 2 (parallel workers) — load edges as datasetUrl, receive node map from previousLayerResultJson, and each run Dijkstra from source nodes WorkerIndex×20 through WorkerIndex×20+19: https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_13_edges.jsonl (u, v, weight_minutes) Return: {top_20_betweenness_nodes, top_20_critical_edges, avg_path_length_hops, diameter_hops}
Formal Definition	On a synthetic weighted undirected road-network graph (N = 5,000 nodes, E = 25,000 edges, weights = travel time in minutes), compute: shortest-path lengths from 400 sampled source nodes (Dijkstra); approximate betweenness centrality for all nodes; network diameter and average path length; and the top-20 most critical edges by removal impact.
• Two-layer pipeline	Graph (5,000 nodes, ~25,000 edges) pre-generated and published on HuggingFace. Node fields: node_id, x, y. Edge fields: u, v, weight_minutes. Two-layer pipeline: Layer 1 (1 worker) loads nodes from task_13_nodes.jsonl and passes the node coordinate map via resultJson; Layer 2 workers each run Dijkstra from 20 source nodes (source indices WorkerIndex×20 ... +19), loading edges from task_13_edges.jsonl as datasetUrl.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_13_nodes · task_13_edges · task_13_sources · benchmark (task_id=13)
Parallelisation	Each worker runs Dijkstra from 20 source nodes and accumulates partial betweenness centrality. A final layer sums centrality vectors globally.
Applicability	Used in urban planning (bottleneck roads), telecom design, and supply chain resilience analysis.
Success Metric	Betweenness centrality ranking, top-20 critical edges, average path length, diameter.
Timing	Sequential (Agent B): ~30 min Parallel 20× (Agent A): ~2 min

• Agent Prompt	Simulate photon transport through a 50-layer plane-parallel atmosphere. Layer l : $\tau_{\text{scat}}[l]=0.1 \cdot e^{-(0.05l)}$, $\tau_{\text{abs}}[l]=0.02 \cdot e^{-(0.08l)}$; $\omega[l]=\tau_{\text{scat}}/(\tau_{\text{scat}}+\tau_{\text{abs}})$. Solar zenith $\theta=30^\circ$; Rayleigh phase function. Simulate 5,000,000 photons total; each worker traces its batch using <code>seed=WorkerIndex·6271+3</code> . Return: <code>{toa_upwelling_radiance, surface_downwelling_irradiance, per_layer_heating_rates:[50 floats], ssa_retrieval_rmse}</code>
Formal Definition	Simulate photon transport through a 50-layer plane-parallel atmosphere. Each photon is launched at solar zenith angle $\theta = 30^\circ$. At each layer, the photon scatters (Rayleigh phase function), is absorbed (absorption optical depth τ_{abs}), or exits. Simulate 5,000,000 photons to estimate TOA upwelling radiance, surface downwelling irradiance, per-layer heating rates, and single-scatter albedo retrieval accuracy.
• Mathematical parameters	No external dataset. Atmospheric profile fully specified: layer optical depths $\tau_{\text{scat}}[l] = 0.1 \cdot \exp(-0.05 \cdot l)$, $\tau_{\text{abs}}[l] = 0.02 \cdot \exp(-0.08 \cdot l)$ for $l = 0 \dots 49$. Single-scatter albedo $\omega = \tau_{\text{scat}} / (\tau_{\text{scat}} + \tau_{\text{abs}})$. Each worker traces 250,000 photons using new <code>Random(WorkerIndex * 6271 + 3)</code> .
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: <code>benchmark (task_id=14)</code>
Parallelisation	Workers each trace 250,000 photon paths (5,000,000 / 20). Exit radiances and layer absorption events are accumulated in a final layer.
Applicability	Used to calibrate satellite-borne radiometers (MODIS, Sentinel) and in climate model parameterisation. Sequential 5 M-photon simulation takes ~30 minutes.
Success Metric	TOA radiance within 1% of benchmark, correct heating rate profile, retrieval accuracy.
Timing	Sequential (Agent B): ~30 min Parallel 20× (Agent A): ~2 min

Task 15	Parallel Genome k-mer Frequency Spectrum	Bioinformatics / Genomics
• Agent Prompt	Compute the $k=8$ frequency spectrum ($4^8=65,536$ k-mers) for each of 20 synthetic genomic sequences and for the combined corpus. For each sequence identify the top-50 over/under-represented k-mers vs. null expectation and any repeat regions (k-mer freq>100 in consecutive 10 kb windows). Each worker processes one sequence (<code>seq_id=WorkerIndex</code>). A final aggregation layer computes Jensen-Shannon divergence between each spectrum and the corpus mean. Sequences (<code>seq_id</code>, <code>length</code>, <code>gc_content</code>, <code>sequence</code>): https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark/resolve/main/data/task_15_sequences.jsonl Pass this URL as <code>datasetUrl</code>. Workers read via <code>File.ReadAllText(input.DatasetPath!)</code>. Return: <code>{js_divergences:[20 floats], corpus_top_10_kmers:[str], frequency_table_checksums:[20 ints]}</code>	

Formal Definition	Compute the complete $k = 8$ frequency spectrum ($4^8 = 65,536$ k-mers) for each of 20 synthetic genomic sequences (length 500,000 bp each) and for the combined corpus. Identify the 50 most over- and under-represented k-mers vs. null expectation, detect repeat regions (k-mers with frequency > 100 in consecutive windows), and compute Jensen-Shannon divergence between each sequence's spectrum and the corpus mean.
• HuggingFace dataset	20 genomic sequences (500,000 bp each) pre-generated and published on HuggingFace. Fields: seq_id (int), length, gc_content, sequence (string). Pass the dataset URL as datasetUrl to run_layer. Each worker processes one sequence (seq_id = WorkerIndex). Layer 2 (1 worker) receives all 20 frequency tables via previousLayerResultJson and computes corpus-level Jensen-Shannon divergences.
• HuggingFace Dataset	https://huggingface.co/datasets/parcs-benchmark/parcs-agent-benchmark Splits: task_15_sequences · benchmark (task_id=15)
Parallelisation	Each worker computes one sequence's full 65,536-entry k-mer table. A final Layer 2 worker merges tables and runs statistical tests.
Applicability	k-mer spectra underlie metagenomic classification, repeat detection, and genome assembly QC. Sequential processing of $20 \times 500,000$ bp sequences takes ~20 minutes.
Success Metric	Frequency tables (verifiable by checksum), top over/under k-mers, JS-divergence per sequence.
Timing	Sequential (Agent B): ~20 min Parallel 20× (Agent A): ~1 min